Synthesis of *endo*-1,2,3,4,5-Pentaphenylbicyclo[2.2.1]hept-2-en-7-one by One-Pot Aldol Condensation and Diels–Alder Reaction

CM3191 • 8 and 15 October 2026 • Instructor: Dr Hoang Truong Giang
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1. Aim

Tetracyclone **1** was prepared by KOH-catalysed double aldol condensation of benzil with 1,3-diphenylacetone in triethylene glycol (TEG).[1,2] It was trapped in the same vial by a Diels–Alder reaction with styrene at 110 °C (Figure 1). Each step was followed by TLC. The product was recrystallised from ethanol/ethyl acetate. Its identity among **3A**, **3B** and **4** was assessed by melting point, IR and ^{1}H NMR spectroscopy and the X-ray crystal structure.[3]

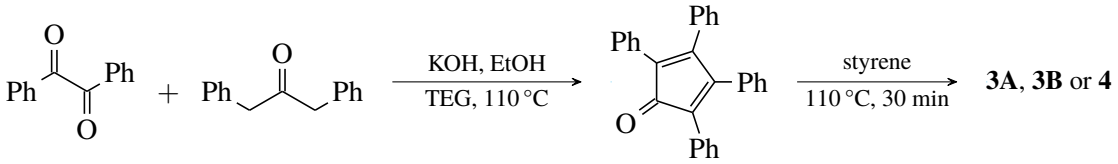


Figure 1. One-pot sequence. **3A** (*endo*) and **3B** (*exo*) are 1,2,3,4,5-pentaphenylbicyclo[2.2.1]hept-2-en-7-one (MW 488.63). **4** is 1,2,3,4,5-pentaphenylcyclohexa-1,3-diene (MW 460.62), formed if CO is lost.[1]

2. Results

Table 1. Reagents.

Reagent	M / g mol^{-1}	Amount	n / mmol	Equiv.
Benzil (limiting)	210.23	0.2080 g	0.989	1.00
1,3-Diphenylacetone	210.27	0.21 g (fixed portion)	1.00	1.01
Triethylene glycol	150.17	0.50 mL	—	solvent
KOH, 2 M in EtOH	56.11	5 drops	—	cat.
Styrene (ρ 0.906 g mL^{-1})	104.15	0.40 mL	3.48	3.52

Table 2. Mass and yield, corrected for residual solvent by ^{1}H NMR (calculation in Appendix B).[5]

	3A (MW 488.63)
Product (bag + product − bag)	1.5135 g − 1.3105 g = 0.2030 g
Product content	99.6 wt%
Corrected mass	0.2023 g
Theoretical yield (0.989 mmol)	0.4834 g
Yield, uncorrected / corrected	42.0% / **41.8%**

The datasheet yield (41.6%) used a nominal 1.00 mmol of benzil.

Table 3. TLC (silica, 10% EtOAc/hexane, UV). Spots are given as lower and upper edge; R_f uses the spot centre.

Plate (front)	Lane	Spot / cm	R_f	Comment
1 (4.5 cm)	Benzil (B)	1.3–1.5	0.31	B and D not resolved
	1,3-Diphenylacetone (D)	1.3–1.5	0.31	
	R1	1.8–2.0	0.42	single spot
2 (4.5 cm)	R1	2.0–2.3	0.48	main spot, streak at 1.5–1.7 cm
	R2	1.8–2.0	0.42	new lower spot, streak at 1.5–1.7 cm

Table 4. Observations.

Stage	Observation
KOH (5 drops)	Turned deep purple, almost black, almost immediately.
Styrene, 110 °C, 30 min	R2 sample bright yellowish brown (café latte to milk tea). TLC at 11:42–11:59 showed completion. No extra styrene added.
Recrystallisation	10 mL EtOH and 10 mL EtOAc, each from a fixed-volume dispenser. The mixture looked white to pale pink while heating. Extra hot EtOAc was added by dropper until the solid dissolved; its volume was not measured. The solution was then clear and bright pink. It was left at room temperature for one week, covered with foil pierced with a few holes. Crystals were filtered in the next session (15 October).
Product	White, slightly translucent crystals; white when ground for the melting point.
Melting point (Mel-Temp)	176.4–187.8 °C. Browned as melting began; black liquid when fully molten.

Table 5. ^{1}H NMR (500 MHz, $CDCl_3$, δ relative to $CHCl_3$ at 7.26)[6] and IR data. H_a is the benzylic CH and H_b/H_c the CH_2.

δ / ppm	Mult.	J / Hz	Int.	Assignment
7.62–6.42	m	—	26.2[a]	25 aromatic H
4.30	dd	10.0, 6.5	1.00	H_a
3.18	dd	12.3, 10.0	0.99	H_b, *cis* to H_a
2.65	dd	12.3, 6.5	0.99	H_c, *trans* to H_a
2.05	s	—	0.04	EtOAc
IR (solid, $\tilde{\nu}$/cm^{-1}) 3048, 3028 (Ar C–H), **1773** (C=O), 1601, 1494, 1439 (Ar C=C), 751, 693 (Ph)				
X-ray structure ($C_{37}H_{28}O$) *endo* adduct **3A**				

[a] Includes residual $CHCl_3$. Groups 7.62–7.58 (2H), 7.47–7.42 (2H), 7.38–7.05 (14H), 6.95–6.88 (3H), 6.81–6.76 (2H), 6.46–6.42 (2H).

3. Discussion

3.1 Reaction monitoring

The mixture turned deep purple, almost black, almost immediately after KOH was added. This colour indicates the conjugated tetracyclone **1**. On plate 1, R1 gave one spot at R_f 0.42. No spot remained at R_f 0.31. Both starting materials were therefore consumed in the aldol step. Benzil and 1,3-diphenylacetone co-eluted, however. Had one remained, the plate could not have shown which.

On plate 2, R2 ran slightly below R1. The colour changed from deep purple, almost black, to yellowish brown. Both changes indicate that **1** was consumed. R1 ran at R_f 0.42 on plate 1 but 0.48 on plate 2. Lanes were therefore compared within one plate. The streak at 1.5–1.7 cm may come from overloading with TEG-containing samples.

3.2 Product identity

The spectrum shows 25 aromatic H and three 1H doublets of doublets. This ratio fits a 1:1 adduct of **1** and styrene. No styrene vinyl signals appear at δ 5.2–5.8.

H_b and H_c share a geminal coupling of 12.3 Hz. H_a couples to them with 10.0 and 6.5 Hz. Together they form a CHPh–CH_2 unit. In a rigid bicyclo[2.2.1] skeleton, *cis* vicinal protons are nearly eclipsed. *Trans* vicinal protons lie near 120°. The Karplus relationship predicts the larger coupling for the *cis* pair.[7] H_b is therefore *cis* to H_a, and H_c is *trans*. The 2H multiplet at δ 6.44 is strongly shielded. A phenyl ring may lie over the face of a neighbouring ring in this crowded framework.

Compound **4** also contains a CHPh–CH_2 unit. The NMR data therefore cannot separate **3A/3B** from **4**. IR settles this point. The IR spectrum shows a strong C=O band at 1773 cm^{-1}. This lies about 60 cm^{-1} above an unstrained ketone near 1715 cm^{-1}. Ring strain raises the C=O frequency.[8] The band fits the strained bridging ketone of **3A/3B**. Compound **4** has no carbonyl group and is ruled out.

Both **3A** and **3B** have one *cis* and one *trans* CH_2 proton relative to H_a. The J values therefore cannot distinguish *endo* from *exo*.

The X-ray structure ($C_{37}H_{28}O$) shows the *endo* adduct **3A**.[3,4] The styrene phenyl lies *anti* to the carbonyl bridge. It sits on the same side as the C3=C4 bridge. The torsion angles confirm the NMR assignment. H6A–C6–C7–H7A is $-12.6°$, close to eclipsed. H6A–C6–C7–H7B is 133.3°. H6A is therefore H_a, H7A is H_b and H7B is H_c (handout numbering). The C2–C1–C5 angle at the carbonyl carbon is only 99.0°. This compression fits the high C=O frequency. The *endo* adduct could have formed under kinetic control, favoured by secondary orbital overlap.[2]

3.3 Melting behaviour

The sample melted at 176.4–187.8 °C. It browned as melting began and ended as a black liquid. The product can release styrene above 170 °C.[1] This retro-Diels–Alder reaction would regenerate the coloured tetracyclone. Loss of CO could also contribute. The broad range probably reflects decomposition. NMR shows 99.6 wt% product, with only traces of EtOAc and EtOH. The hot recrystallisation solution was clear and bright pink. This colour suggests a trace of intensely coloured **1**. The crystals were white, so this impurity probably stayed in the mother liquor.

3.4 Yield loss

The corrected yield of **3A** was 41.8%. TLC of R2 showed complete conversion. Most of the loss probably occurred during work-up. The solution stood for one week before filtration. Crystallisation time therefore probably did not limit recovery. It was filtered at room temperature without ice cooling. The product remains partly soluble in EtOH/EtOAc at room temperature. Some therefore stayed in the mother liquor. Extra hot EtOAc was added by dropper and may have increased this loss, but its volume was not measured. More was lost in transfers and in the ethanol washes. The crude solid was not weighed. Reaction and purification losses cannot be separated.

3.5 Improvements

Weighing the crude solid would separate reaction and purification losses. Adding the extra EtOAc from a graduated pipette, and keeping it to a minimum, might raise recovery. So might cooling the flask in ice before filtration. A second crop from the concentrated mother liquor could add product, with its purity checked by TLC.[9] These changes would need testing. A less polar eluent or a co-spot could resolve the two starting materials. A sealed capillary might limit decomposition during the melting point.

3.6 Green aspects

The aldol step loses only two molecules of water.[10] Its atom economy is

$$\mathrm{AE}(\mathbf{1}) = \frac{384.48}{210.23 + 210.27} \times 100\% = 91.4\%$$

The Diels–Alder step incorporates all atoms of both partners. Over both steps,

$$\mathrm{AE}(\mathbf{3}) = \frac{488.63}{210.23 + 210.27 + 104.15} \times 100\% = 93.1\%$$

The one-pot design avoided isolating and redissolving tetracyclone.

4. Conclusion

Tetracyclone was formed by double aldol condensation and trapped in situ with styrene. TLC showed complete conversion in both steps. The recrystallised product, white crystals, weighed 0.2030 g (41.8% after NMR correction). It decomposed on melting at 176–188 °C. ^{1}H NMR showed a 1:1 adduct with a rigid CH–CH_2 unit. The IR C=O band at 1773 cm^{-1} identified the bridged ketone **3A/3B**. The X-ray structure identified the product as the *endo* adduct **3A**.

References

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[3] Hoang, G. T.; Kubo, T.; Young, V. G.; Kautzky, J. A.; Wissinger, J. E. Illustrating the Utility of X-ray Crystallography for Structure Elucidation through a Tandem Aldol Condensation/Diels–Alder Reaction Sequence. *J. Chem. Educ.* **2015**, *92*, 1381–1384.

[4] Rohde, G. T.; Young, V. G., Jr. *X-ray Crystal Structure Data for the Major Product Isolated from the Diels–Alder Reaction of Tetraphenylcyclopentadienone and Styrene* (CM3191 handout, Hoang, T. G.); University of Minnesota; National University of Singapore, 2026.

[5] *Correcting Compound Mass and Yield for Residual Solvent Using ^{1}H NMR* (CM3191 handout); National University of Singapore, 2026.

[6] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[7] Karplus, M. Contact Electron-Spin Coupling of Nuclear Magnetic Moments. *J. Chem. Phys.* **1959**, *30*, 11–15.

[8] Pavia, D. L.; Lampman, G. M.; Kriz, G. S.; Vyvyan, J. R. *Introduction to Spectroscopy*, 5th ed.; Cengage Learning: Stamford, CT, 2015.

[9] Furniss, B. S.; Hannaford, A. J.; Smith, P. W. G.; Tatchell, A. R. *Vogel's Textbook of Practical Organic Chemistry*, 5th ed.; Longman: Harlow, 1989.

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Data and Appendix

A Signed datasheet

B Residual solvent and yield[5]

Integrals are relative to H_a (1H) = 1.00. The EtOAc singlet at δ 2.05 (3H) integrates to 0.0433. The EtOH quartet at δ 3.73 (2H) integrates to 0.0203.

$$N_{\text{EtOAc}} = \frac{0.0433}{3} = 0.01443 \qquad N_{\text{EtOH}} = \frac{0.0203}{2} = 0.01015 \qquad N_{\mathbf{3}} = 1.000$$

$$\text{wt\%}_{\mathbf{3}} = \frac{1.000 \times 488.63}{1.000 \times 488.63 + 0.01443 \times 88.11 + 0.01015 \times 46.07} \times 100\% = 99.645\%$$

$$m_{\text{pure}} = 0.2030\ \text{g} \times 0.99645 = 0.20228\ \text{g}$$

$$n_{\text{benzil}} = \frac{0.2080\ \text{g}}{210.23\ \text{g mol}^{-1}} = 0.98939\ \text{mmol} \qquad m_{\text{theor}} = 0.98939\ \text{mmol} \times 488.63\ \text{g mol}^{-1} = 0.48345\ \text{g}$$

$$\text{Yield} = \frac{0.20228\ \text{g}}{0.48345\ \text{g}} \times 100\% = 41.84\% \approx 41.8\%$$

C ^{1}H NMR spectrum (data set CM3191-Thurs-5TBI-Oct15-2026-One Pot-09)

D IR spectrum (KIMDONGHWAN_Experiment 2, 15 October 2026)

E X-ray crystal structure[4]

Name: LIM JONGHWAN

Date of the experiment: 2026/10/08/Th. (synth.)
2026/10/15/Th. (characterisation)

Datasheet for One pot Aldol Condensation and Diels-Alder Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

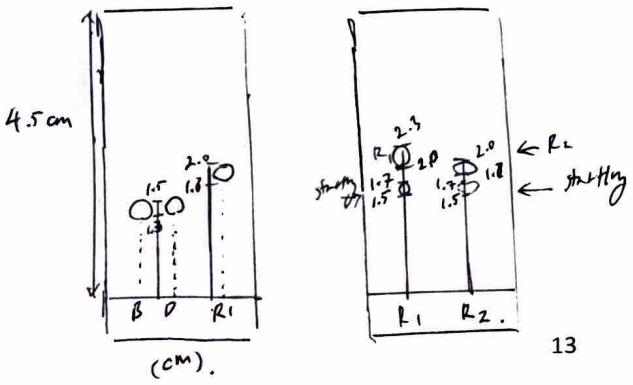
- Actual mass of benzil used (g) 0.2080
- Actual mass of 1,3-diphenylacetone used (g) fixed. (fwd pipetting).
- Amount of styrene used (mL) 0.4 (bottom minisc., eye level).

Product

- Mass of empty plastic bag (g) 1.3105 g.
- Mass of plastic bag + product (g) 1.5135 g.
- Mass of your final product (g) 0.2030 g.
- Percentage yield (%): 41.6 ~ 44.1 %.

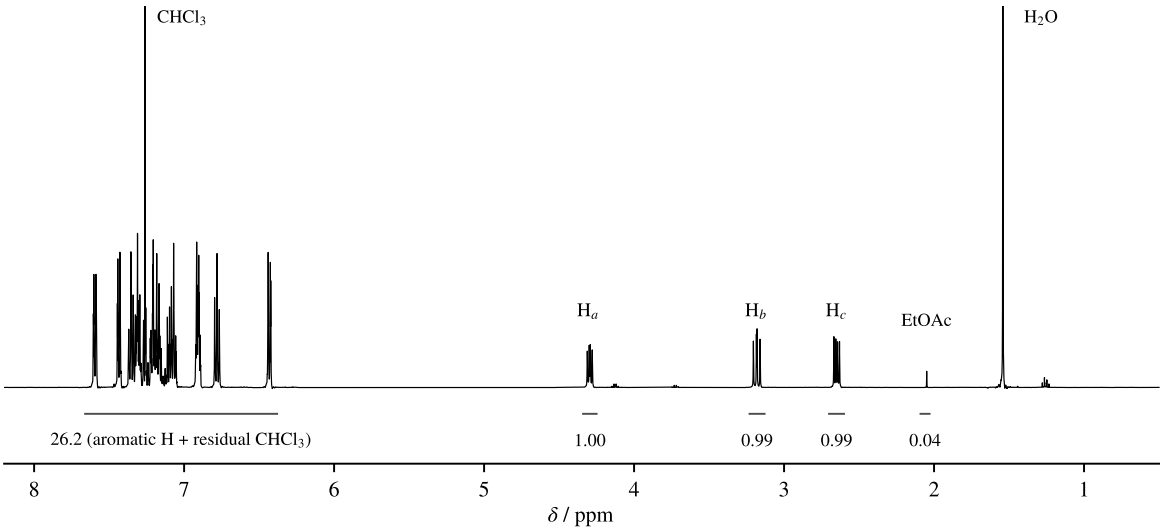
Observation during the experiment: start – end.
176.4 – 187.8 °C
started melting, brownish
fully molten, black liquid.

Draw your TLC plate here and label all the distances (in cm)

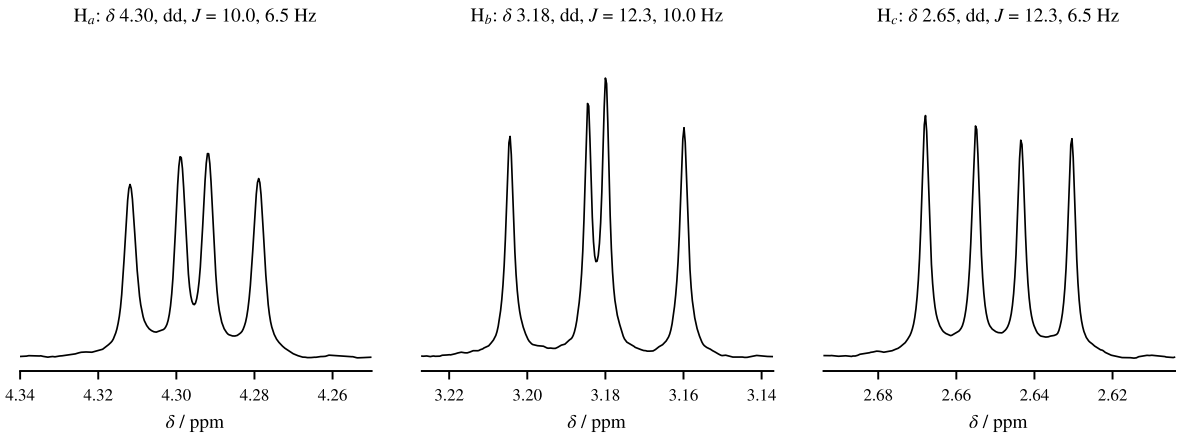


R2 heating
30 min
↓
TLC 11:42
↓ 5
11:59
continues to be finished
bright yellowish brown
(coffee-latte ~ milk tea color)

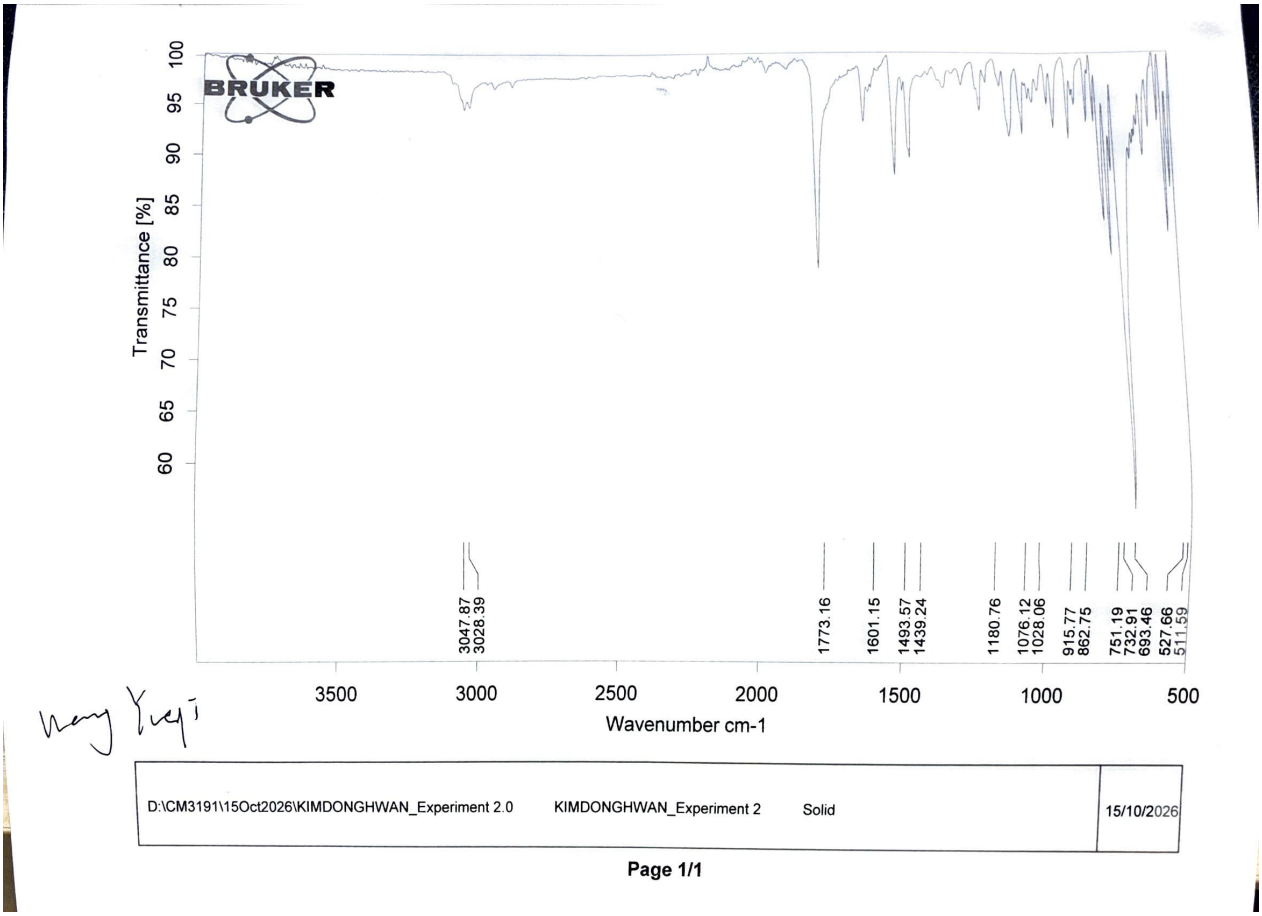
^{1}H NMR (500 MHz, $CDCl_3$, 298 K) of the recrystallised product; δ referenced to $CHCl_3$ (7.26); integrals relative to H_a



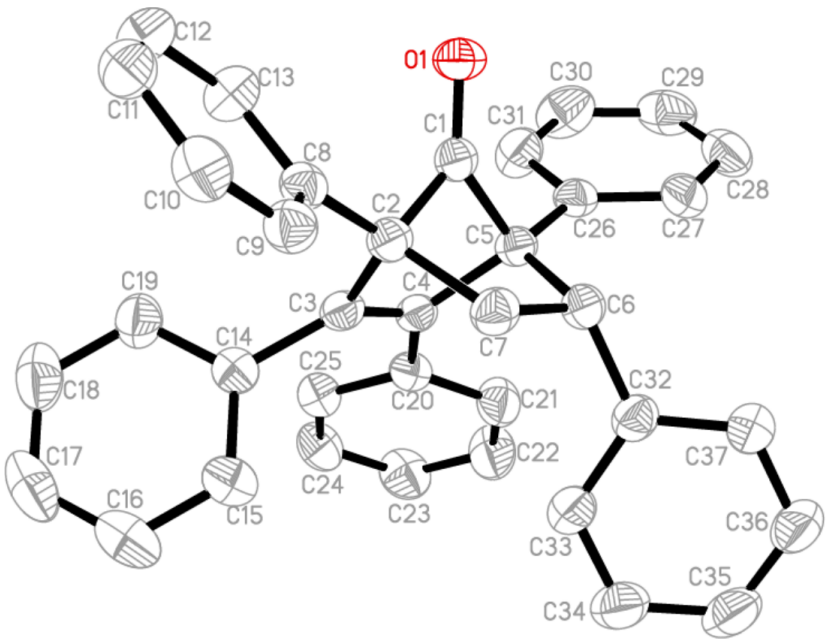
Expansions of the three non-aromatic signals



C ^{1}H NMR spectrum of the recrystallised product (500 MHz, $CDCl_3$).



D IR spectrum of the recrystallised product (solid).



E Thermal ellipsoid plot of the major product, 50% probability ellipsoids (handout numbering).[4] Selected torsion angles H6A–C6–C7–H7A −12.60° and H6A–C6–C7–H7B 133.30°.